Vedant Desai

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EDUCATION

University of Michigan — Expected May 2028
B.S. in Computer Science and Economics — Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

CaseStudyPrep.AI — Dec 2025 – May 2026
Software Engineer Co-op (Voice AI) — Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Michigan Data Consulting (MDC) — Jan 2026 – May 2026
Data Engineer — Michigan Campaign Finance Network — Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Architected a deterministic aggregation engine that parses those exports, normalizes irregular contribution amounts, and ranks PACs by total funding volume so researchers stop rebuilding spreadsheets to surface top spenders.

Vylet — May 2026 – Present | vyletdata.com
Founder — Automated lead-sourcing platform for PE/search-fund; live product generating $1,500 MRR across three clients

- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.
- Engineered Node 3 as a pure-Python triangulated consensus gate — no LLM calls — that computes a 0–100 lead score by fuzzy-matching the pipeline query, state business registry, and live website crawl in a three-way weakest-link check, then hard-fails leads on legal status, industry, geography, or independence before the score threshold applies.

PROJECTS

Granular Synthesizer Plugin | *C++, JUCE* — github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a MemoryPool<Grain, 64> slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so processBlock() never calls new or delete after prepareToPlay() completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic shared_ptr swap so the UI thread never blocks processBlock().
- Built the granular engine around a fractional-accumulator scheduler that fires grains at density × (1 - overlap × 0.5) effective rate — each grain reads a random WAV offset at $2^{(\text{semitones}/12)}$ playback speed, windowed by a pre-computed 8,192-entry Gaussian LUT to eliminate click artifacts at grain edges, with all grain memory drawn from a per-voice pre-allocated pool.

SignalWeaver | *Python, LoRA* — github.com/Verdent06/SignalWeaver

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.

TECHNICAL SKILLS

Languages .... Python, TypeScript, C++, SQL
Frameworks .... Angular, RxJS, JUCE
AI / ML .... Pandas, ONNX Runtime, Silero VAD